

- 1 The diagram shows parallelogram $EFGH$.

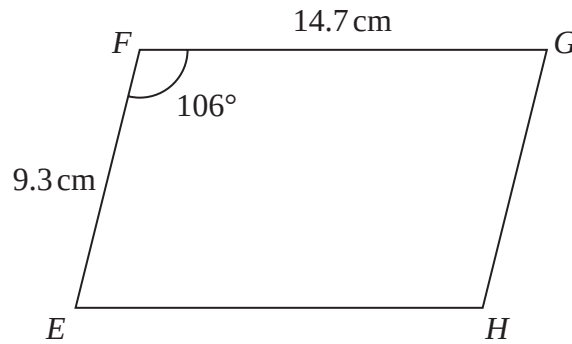


Diagram **NOT**
accurately drawn

$EF = 9.3\text{ cm}$
 $FG = 14.7\text{ cm}$
 Angle $EFG = 106^\circ$

- (a) Work out the area of the parallelogram.
 Give your answer correct to 3 significant figures.

..... cm^2
 (2)

- (b) Work out the length of the diagonal EG of the parallelogram.
 Give your answer correct to 3 significant figures.

..... cm
 (3)

(Total for Question 1 is 5 marks)

2 Here is a right-angled triangle.

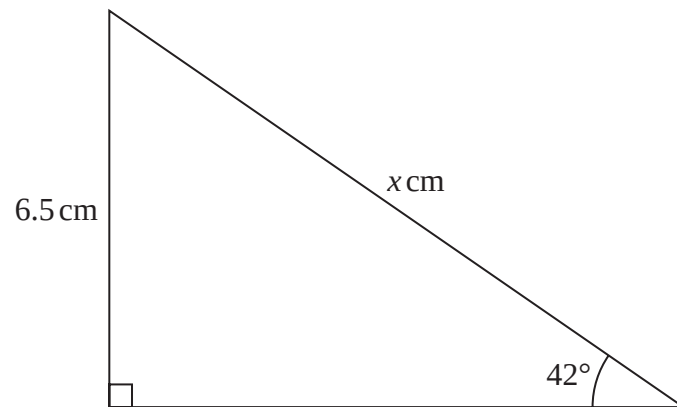


Diagram **NOT**
accurately drawn

Work out the value of x .
Give your answer correct to one decimal place.

$x =$

(Total for Question 2 is 3 marks)

3

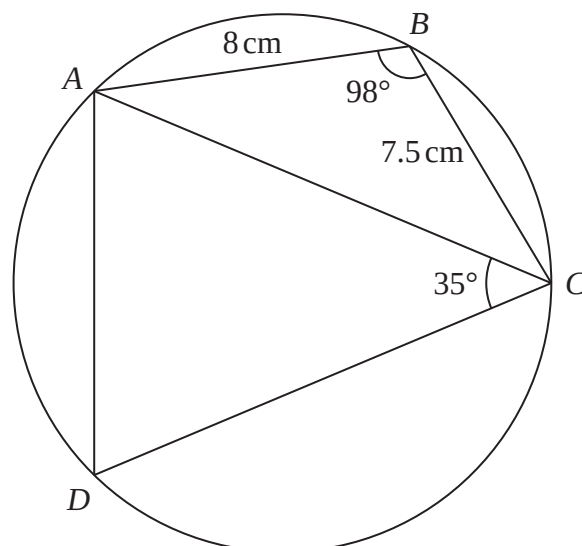


Diagram **NOT**
accurately drawn

$ABCD$ is a quadrilateral where A , B , C and D are points on a circle.

$$AB = 8\text{ cm}$$

$$BC = 7.5\text{ cm}$$

$$\text{Angle } ABC = 98^\circ$$

$$\text{Angle } ACD = 35^\circ$$

Work out the perimeter of quadrilateral $ABCD$.

Give your answer correct to one decimal place.

..... cm

(Total for Question 3 is 6 marks)

- 4 The diagram shows the positions of three ships, A , B and C .

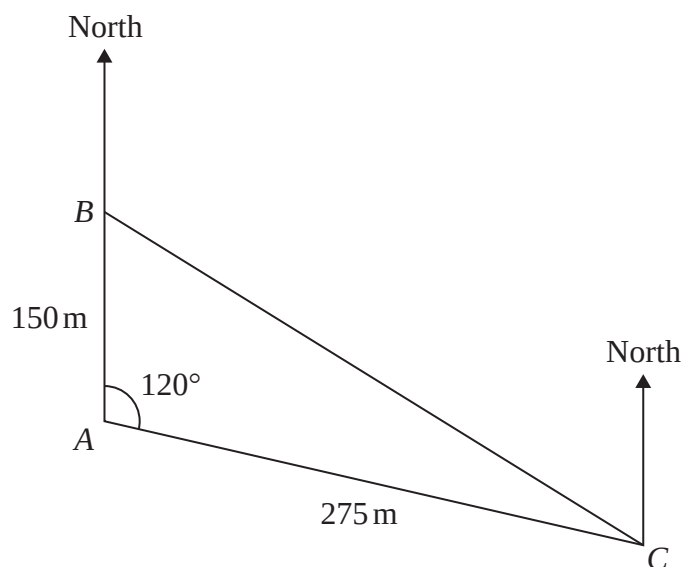


Diagram **NOT**
accurately drawn

Ship B is due north of ship A .

The bearing of ship C from ship A is 120°

Calculate the bearing of ship C from ship B .

Give your answer correct to the nearest degree.

(Total for Question 4 is 5 marks)

- 5 The diagram shows a sector OBC of a circle with centre O and radius $(6 + x)$ cm.

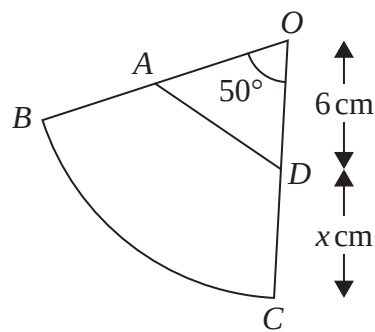


Diagram **NOT**
accurately drawn

A is the point on OB and D is the point on OC such that $OA = OD = 6$ cm

Angle $BOC = 50^\circ$

Given that

the perimeter of sector $OBC = 2 \times$ the perimeter of triangle OAD

find the value of x .

Give your answer correct to 3 significant figures.

$$x = \dots\dots\dots$$

(Total for Question 5 is 6 marks)

6 Here is triangle ABC

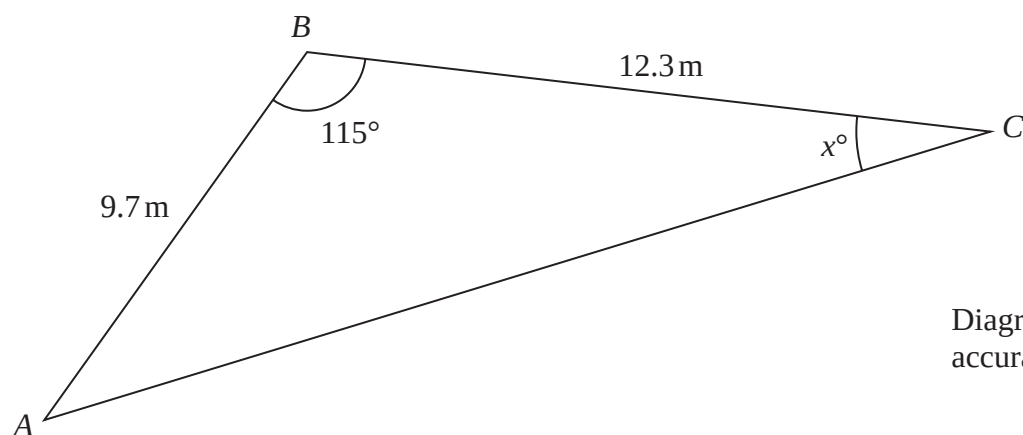


Diagram **NOT**
accurately drawn

Work out the value of x
Give your answer correct to 3 significant figures.

$x =$

(Total for Question 6 is 5 marks)

7 The diagram shows quadrilateral $ABCD$

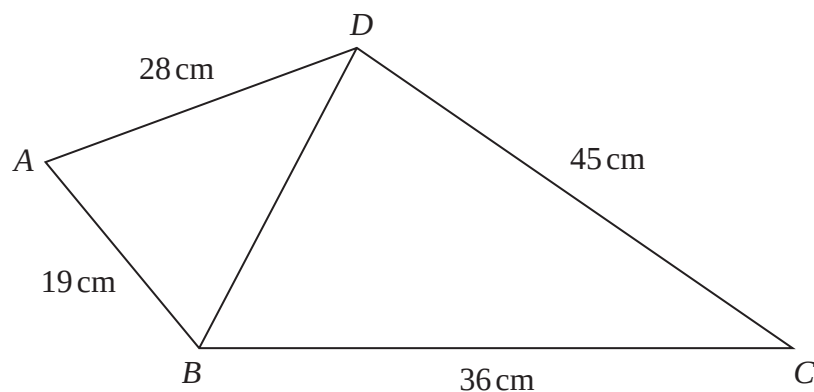


Diagram **NOT**
accurately drawn

The angle BCD is acute.

Given that the area of triangle $BCD = 405\text{ cm}^2$

work out the size of angle ABD

Give your answer correct to one decimal place.

(Total for Question 7 is 5 marks)

8 The diagram shows triangle ABC

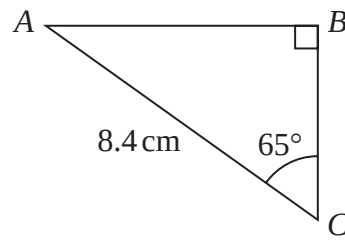


Diagram **NOT**
accurately drawn

Work out the length of the side AB

Give your answer correct to 3 significant figures.

..... cm

(Total for Question 8 is 3 marks)

9 The diagram shows triangle PQR

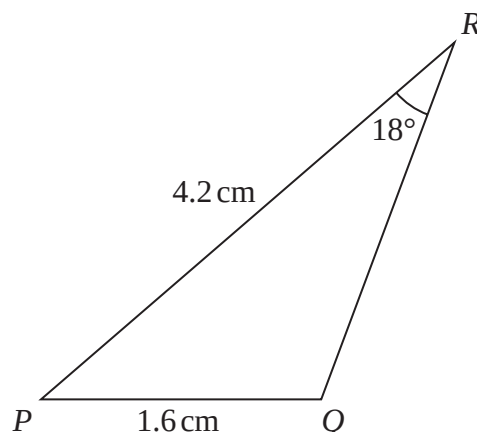


Diagram **NOT**
accurately drawn

$$PQ = 1.6 \text{ cm}$$

$$PR = 4.2 \text{ cm}$$

$$\text{Angle } PRQ = 18^\circ$$

Given that angle PQR is obtuse,

work out the area of triangle PQR

Give your answer correct to 3 significant figures.

..... cm^2

(Total for Question 9 is 6 marks)

10 AEC and BED are chords of a circle.

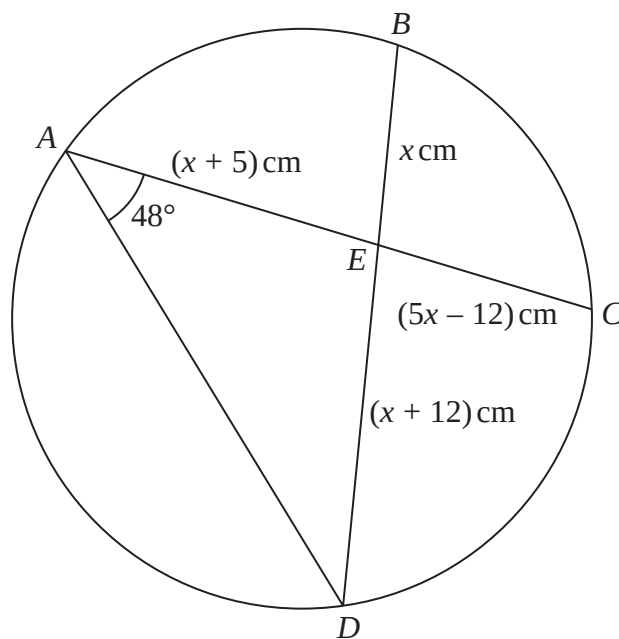


Diagram **NOT**
accurately drawn

$$AE = (x + 5) \text{ cm} \quad BE = x \text{ cm} \quad CE = (5x - 12) \text{ cm} \quad DE = (x + 12) \text{ cm}$$

$$\text{Angle } DAE = 48^\circ$$

Work out the size of angle ADE

Give your answer correct to one decimal place.

o

(Total for Question 10 is 5 marks)

- 11 Here is a shape formed from two triangles ABC and CDE
 ACD and BCE are straight lines.

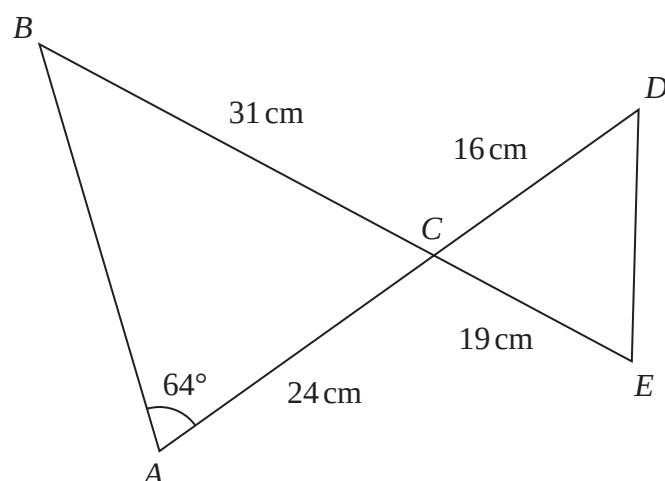


Diagram **NOT**
accurately drawn

$$AC = 24 \text{ cm} \quad BC = 31 \text{ cm} \quad CE = 19 \text{ cm} \quad CD = 16 \text{ cm}$$

$$\text{Angle } BAC = 64^\circ$$

Work out the length of DE

Give your answer correct to 3 significant figures.

..... cm

(Total for Question 11 is 5 marks)

12 Here is a triangle ABC

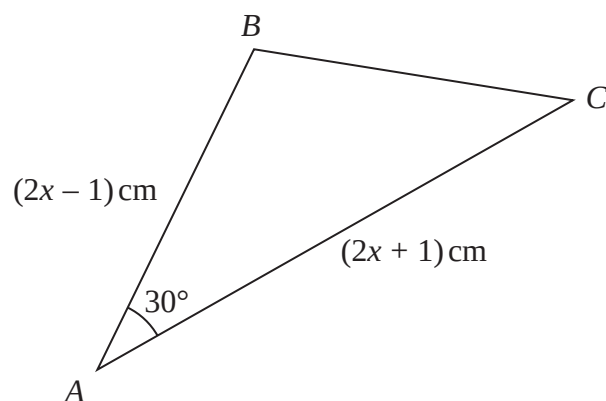


Diagram **NOT**
accurately drawn

The area of the triangle is $(x^2 + x - 3.75) \text{ cm}^2$

Find the size of the largest angle in triangle ABC

Give your answer correct to the nearest degree.

o

(Total for Question 12 is 6 marks)

13 The diagram shows a triangle ABC and a flagpole BF

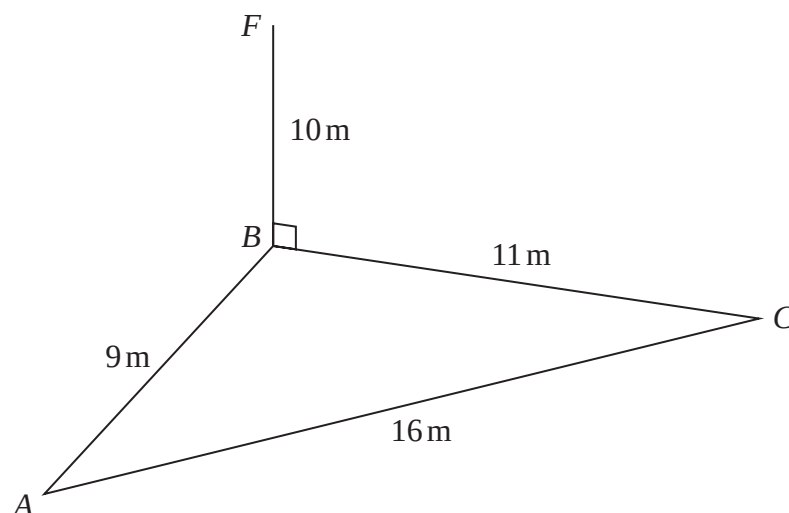


Diagram **NOT**
accurately drawn

A , B and C are points on horizontal ground.

BF is vertical.

$$AB = 9\text{ m} \quad BC = 11\text{ m} \quad AC = 16\text{ m} \quad BF = 10\text{ m}$$

D is the point on AC such that angle $BDC = 90^\circ$

Work out the size of the angle of elevation of the point F from the point D
Give your answer correct to one decimal place.

o

(Total for Question 13 is 5 marks)
